

Instrumentation systems engineer: sensor/probe diagnostics, signal-chain calibration (CEM/ESA/MCA), DAQ automation (PyVISA/Keithley), ADC/FPGA readout, and vacuum/HV test stands; targeting space instrumentation, lab measurement, and experimental systems roles.

EDUCATION

University of Michigan Master of Engineering, Space Systems Engineering GPA: 3.79/4.0 Vellore Institute of Technology (VIT) Bachelor of Technology, Mechanical Engineering GPA: 7.78/10	Ann Arbor, MI, USA Aug 2024–Dec 2025 Vellore, India Jul 2019–Jul 2023
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TECHNICAL SKILLS

Instrumentation, Calibration & Diagnostics: CEM/ESA calibration, charge-injection & MCA signal-chain characterization (gain, centroid, FWHM), RPA / emissive-probe / Langmuir-probe diagnostics, ion-optics energy-angle scans, Amptek DPPMCA, SIMION, SRIM

DAQ, Controls & Lab Automation: Python, LabVIEW (CLAD prep), MATLAB (learning), GPIB/IEEE-488, USB-serial, NI VISA (PyVISA, PyMeasure), Keithley DAQs and SMUs, TDK Lambda PSUs, HV switching, synchronized PSU/DAQ logging

Digital Readout & FPGA: ADC requirement sizing, Zmod ADC 1410, Zynq-7000 / Eclipse Z7 path, AMD Vivado, MATLAB HDL Coder & DSP Toolbox, Verilog/VHDL (learning), FPGA design workflow

Vacuum / HV Test Stands & Mechanical: Vacuum chamber operation (CTI-Cryogenics Hi-Vac; roughing to ~1 mTorr), 1.3 kV HV interface validation, SolidWorks, Fusion 360, AutoCAD, Inventor, ANSYS (Fluent/Mechanical), OpenFOAM

INSTRUMENTATION & MEASUREMENT EXPERIENCE

Solar and Heliospheric Research Group, University of Michigan Student Research Assistant, SSD Instrumentation Readout Chain	Ann Arbor, MI, USA Jan 2025–Dec 2025
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- Sized ADC bit-depth and sampling-rate requirements from energy-resolution targets, CSA/ENC noise limits, and a 25 keV charge case for a solid-state detector readout chain
- Architected a digitization upgrade path from 1 MSPS to 125 MSPS by integrating a 14-bit Zmod ADC 1410 into a Zynq-7000 / Eclipse Z7 programmable-logic architecture
- Verified MATLAB HDL Coder golden models against Vivado co-simulations and diagnosed 125 MHz post-synthesis timing limits (Worst Negative Slack) in the Zynq SoC signal path

Climate and Space Sciences and Engineering, University of Michigan Research Assistant, LVACCS Instrumentation & Test-Stand Characterization	Ann Arbor, MI, USA Feb 2026–Present
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- Built an instrumented HV test workflow (1.3 kV) with discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data capture
- Operated and interpreted RPA, emissive-probe, and Langmuir-probe instrumentation to compare chamber conditions across electrostatic regolith-lofting test states
- Automated remote ignition, Keithley DAQ sequencing, and post-run processing for Spacecraft Charging Device readiness, improving repeatability and saving ~15 minutes of processing per run

Space Physics Research Lab, University of Michigan Summer Intern, TestBedz Spacecraft Qualification Platform	Ann Arbor, MI, USA May 2025–Jul 2025
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- Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility instrumentation limits across 6 test-type workflows
- Encoded hardware metadata, facility operating envelopes, and acceptance criteria into machine-readable compatibility checks and traceable test plans for repeatable qualification workflows

L3Harris and University of Michigan Communications Lead, Uranian Orbiter and Probe Mission Study	Ann Arbor, MI, USA Sep 2024–May 2025
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- Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% (\$450M) cost reduction through an orbiter-only configuration trade

SELECTED INSTRUMENTATION PROJECTS

Space Instrumentation Calibration & Ion-Optics Series <i>Graduate Course: SPACE 571 Supervisor: Prof. Stefano Livi</i>	Jul 2025–Dec 2025
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- Calibrated Channel Electron Multiplier (CEM) response by mapping operating bias, pulse-height distributions, and amplifier gain with charge-calibrated PocketMCA readout; linked detector settings to signal quality and count-rate interpretation
- Characterized the end-to-end signal chain—charge injection → CSA / Charge-Shaper Amplifier → pulse shaping → MCA acquisition—extracting gain, centroid, FWHM, and bias-dependent count response
- Calibrated Cylindrical Electrostatic Analyzer (ESA) performance from lab energy-angle scans and SIMION/SRIM ion-optics models, quantifying transmission, angular acceptance, energy resolution, and filter response

PROJECTS & LEADERSHIP

CANSAT 2022 (NASA & AAS) Ranked 7th worldwide out of 42 teams	Aug 2021–Jul 2022
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- Built a 10 m tether-deployment, payload release, and camera-stabilization system using a DC motor, worm gear, torsion-spring latch, and dual-servo gimbal for controlled descent and fixed 45° imaging

Spaceport America Cup / IREC Placed 23rd globally and 5th in Asia-Pacific	Apr 2020–Jul 2021
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- Launched a Mach 0.9 solid-motor rocket to 10,000 ft and developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction